

Ahsan Tahir

AI Engineer

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PROFILE

AI/ML Engineer with 2+ years of production experience specializing in Computer Vision, LLM engineering, and MLOps pipelines. Expert in training and deploying YOLO architectures, building advanced RAG networks using LangChain/LangGraph, and Gemini APIs. Proven track record of taking complex machine learning models out of research notebooks and scaling them into production-ready FastAPI environments via Docker and VPS.

PROFESSIONAL EXPERIENCE

Python/AI Engineer

05/2024 – present

Concave Tech

AI-Powered Clustering & Route Optimization System:

A full-stack web application that intelligently clusters geographic locations and optimizes daily shop visit schedules. The system uses advanced machine learning algorithms to:

- **Intelligent Geographic Clustering:** Uses constrained K-means clustering to group shops by location while respecting business constraints (maximum visits per cluster, distance limits from cluster centers, time availability).
- **Daily Schedule Optimization:** Distributes shop visits across multiple days while balancing workload, maintaining visit frequency requirements, and maximizing time utilization.
- **Route Optimization:** Devised algorithms for geodesic distances between locations and optimized travel routes using algorithms like heap-based path finding.
- **Enterprise Features:**
 - User authentication and session management with MySQL database.
 - Excel file import/export for bulk data handling.
 - Interactive web dashboard with real-time data visualization.
 - REST API with CORS support for external integrations.
 - Performance-optimized processing using Numba JIT compilation.

Retail Planogram Compliance System:

- Engineered a computer vision pipeline using **YOLO & Geometric Algorithms** to automate shelf compliance checks (Horizontal/Vertical matching).
- Improved complex logic for **Brand Blocking** detection and **Eye-Level (Golden Zone)** analysis, directly impacting retail audit efficiency.
- Architected an interactive audit dashboard using **Gradio** for visualizing gaps and misplacements

Real-Time Object Detection API & MLOps:

- Cleaned, validated, and structured large-scale datasets for model training, ensuring data quality and consistency.
- Built and deployed a production-grade **FastAPI** backend for real-time inference.
- **Containerized** applications using **Docker** for containerization, handling image builds, tagging, and server-side deployment on Linux environments.
- streamlined large-scale datasets using **AWS CLI** with S3-compatible object storage (Wasabi).

EDUCATION

University of the Punjab

2019 – 2023

Bachelor of Science in Information Technology (BS-IT)

PROJECTS

ShelfVision AI – Retail Compliance Analyzer 📄

- Built a production AI system that combined Computer Vision and LLM engineering. The Gemini Vision API analyzes shelf images for planogram compliance with structured JSON output.
- Implemented a RAG pipeline using a ChromaDB vector database to store and retrieve audit history. This enabled trend analysis across multiple store audits.
- **Orchestrated** a LangGraph React agent with 4 custom analytical tools for autonomous compliance trend calculation and report generation.
- Developed a FastAPI backend with 5 REST endpoints and a React + Vite + Tailwind CSS frontend. This was deployed on Hugging Face Spaces.

SKILLS

LLM & AI: OpenCV, NLP, LangChain, **LangGraph**, RAG, ChromaDB, Gemini Vision API

Frameworks & Libraries: Pandas, PyTorch, **YOLO**, TensorFlow Lite, PyQt, Scikit-learn, Gradio

Cloud & DevOps: VPS, Docker, Git, Wasabi, Hugging Face

Databases: MySQL, SQLite, Firebase, ChromaDB

Backend: FastAPI, Flask

Languages: SQL, Python, Java, JS, C++

Web Technologies: HTML5, CSS3, Angular, React